

ALI RAZA

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AI Research Engineer · Medical AI · Generative Models · Computer Vision

RESEARCH INTERESTS

Medical Image Analysis · Vision-Language Models · Generative Models (GANs, Diffusion, Flow Matching) · Temporal Trajectory Modeling · Domain Adaptation · Explainable Clinical AI · MRI Reconstruction

EDUCATION

FAST-NUCES, Faisalabad, Pakistan

2022 – 2026

BSc Computer Science GPA: 2.94

Thesis: RadGuard — Multimodal Error Detection in AI-Generated Radiology Reports

PUBLICATIONS & ABSTRACTS

- Khan MM, **Raza A**, et al. “Generalizable Detection of CIEDs Using a Hybrid Transformer and CNN Framework.” *ACR 2026 — American College of Radiology Annual Meeting*, Washington DC. [Accepted]
- **Raza A**. “Inter-Lobe Boundary Channels and Localised Folding Repair for Expiration to Inspiration Lung CT Registration.” *Learn2Reg 2026 Challenge, MICCAI 2026*, Strasbourg. Top-4 finalist (validation phase); invited oral presentation and poster. [Under Review]

RESEARCH EXPERIENCE

RadGuard: Multimodal Error Detection in AI-Generated Radiology Reports

2025 – 2026

Final Year Thesis · FAST-NUCES BioViL-T · CXR-BERT · MIMIC-CXR · FastAPI · Kubernetes · AWS EC2

- Designed multimodal VLM detecting 4 error classes (Supported, Hallucinated, Missing, Inaccurate) in AI-generated CXR reports — **Val F1=0.66** on 74,060-sample MIMIC-CXR dataset across 14 conditions.
- Proposed **ELRRS** (Error-Level Radiology Report Score): novel weighted reliability metric via multimodal fusion of ResNet-50 + CheXBERT aligned with ACR + RadGraph standards.
- Implemented **Bidirectional Cross-Attention** (BioViL-T + CXR-BERT); multi-task learning with pseudo-mask supervision; visual explainability via **GradCAM + CBAM**; FAISS-based RAG chatbot for diagnostic querying.

Learn2Breath: Deformable Chest CT Registration — MICCAI 2026 Learn2Reg Challenge

2026

Registration · NLST Cohort · ConvexAdam · TotalSegmentator · MIND-SSC · Dense Displacement Fields

- Built optimisation-based deformable registration combining TotalSegmentator lobe/fissure channels with MIND-SSC features in a ConvexAdam pipeline; achieved **Lobar DSC 0.973, %NDV 0.0035** on validation.
- Designed a Jacobian-based folding repair stage eliminating foldings at near-zero accuracy cost (DSC cost -0.00025); ranked **top-4 (3rd place)** in the Learn2Reg 2026 validation phase.
- Invited to give an **oral presentation** and poster at Learn2Reg 2026 (MICCAI 2026, Strasbourg); invited to submit a full paper to the *Journal of Medical Imaging*.

ProgressGAN-CXR: Temporally Conditioned GANs for Disease Progression Simulation

2025 – 2026

DCGAN · WGAN-GP · Spectral GAN · Knowledge Distillation · PyTorch · FastAPI · React · HuggingFace

- Introduced continuous severity conditioning into GAN-based CXR synthesis for smooth disease trajectory simulation (Normal → COVID-19); demonstrated **FID and Spearman r are orthogonal metrics** requiring separate evaluation.
- Knowledge distillation from WGAN-GP teacher resolved quality-semantics tradeoff: KD Generator achieves **FID 143** matched to DCGAN while improving **Spearman r to 0.984**; DCGAN augmentation achieved **perfect recall (1.00)** on rarest class.
- Deployed as full production application: React frontend (Vercel), FastAPI backend (HF Spaces), 4 live model architectures, Prometheus metrics. Live: progressgan-cxr.vercel.app

E2E-VarNet MRI Reconstruction: Uncertainty-Based Drift Signal Under Distribution Shift

2025–2026

Reproduction + Extension · Sriram et al. MICCAI 2020 PyTorch · fastMRI · Volume-Level Block Bootstrap · Kaggle/Vast.ai

- Reproduced E2E-VarNet for 4x-accelerated single-coil knee MRI reconstruction (T=4/6/8 cascade depths, SSIM 0.7594/0.7606/0.7607 volume-level knee, 0.6622/0.6705/0.6773 brain OOD) and rigorously tested three research extensions with volume-level clustering, bootstrap CIs, and trivial-baseline controls.
- **Headline result:** checkpoint-ensemble uncertainty magnitude rises reliably under knee→brain distribution shift — **1.54x, bootstrap 95% CI [1.35x, 1.74x]** — computable on 100% of tested volumes (vs a competing eSNR signal computable on only 12/15), giving a production-usable drift signal.
- Ran two independent rounds of statistical re-analysis on my own results — corrected pseudoreplication (slice-level → volume-level block bootstrap), documented 19 debugging findings (coil-combination phase cancellation, Adam epsilon-dominance zeroing TTA gradients, mask-seed non-determinism), and reported cascade-depth-vs-OOD and self-supervised TTA extensions as honest near-null and controlled-negative results.

CIED Detection Pipeline — ACR 2026 Accepted Abstract

2024 – 2025

DINOv3 · DenseNet-121 · TransU-Net · nnU-Net · SAM · Vision Transformers

- Three-stage pipeline: DINOv3-based segmentation isolates device; hybrid dual-branch classifier identifies manufacturer (CLS + pooled patch tokens, 1536d); hierarchical model classifier conditioned on manufacturer embedding (64d prior).

- Handles class imbalance via FocalLoss, WeightedRandomSampler, Mixup, and SMOTE in hybrid feature space; generalizes across unseen device morphologies.

TECHNICAL SKILLS

Medical AI: BioViL-T, CXR-BERT, BioViT, CheXBERT, TransU-Net, U-Net, nnU-Net, SAM, DINOv3, ViT
Generative Models: GANs, VAEs, Diffusion, Flow Matching, Knowledge Distillation, Score AND Energy-Based Models
CV & NLP: YOLO (v8/v11), ResNet, DenseNet, GradCAM, CBAM, BERT, Transformers, LoRA/QLoRA, RAG, FAISS
Stack: PyTorch, FastAPI, React, Docker, Kubernetes, AWS EC2, HuggingFace, Vast.ai, Git, Power BI
Datasets: MIMIC-CXR (PhysioNet approved), NIH ChestX-ray14, MS-CXR, CheXpert, FastMRI

INDUSTRY & APPLIED AI EXPERIENCE

Freelance AI / ML Engineer | HusQuay Solutions 2023 – Present
Production AI Deployment · Voice AI Agents · Multimodal Systems · RAG Pipelines

- Deployed GPU-backed **YOLO multimodal detection system** (FastAPI + Vast.ai + React/Vercel) for automated asset depreciation detection in production.
- Built **Voice-Driven B2B Lead Gen Agent** (LiveKit STT/TTS, n8n, MCP) and **Voice-Cloned PDF RAG Assistant** (~80% cloning accuracy) — all production-deployed.

Data Analyst Intern | TAKMIL 2024
Louisville, Kentucky, USA · Remote

- EDA on 100K+ row educational datasets; **Power BI** dashboards; pipeline optimizations improving efficiency by **30%**.

SERVICE & ACTIVITIES

- **Co-Head Hackathon** — ACM Society, Daira 2025 (organized and led university-wide AI hackathon)
- **Member**, ACM Society Management Team — Daira 2024